Editing sample

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The following editing sample is a revision of a technical description written for a college's website and includes my revised text followed by the original text. This document was an assignment from the Technical Communication I course through UC Berkeley Extension. The text required substantial revisions, focusing on structure, transitions, and informative headings.

REVISED TEXT

3D printing: What it is, how it works, and what it's used for

3D printing is a technology that enables you to create a three-dimensional object layer by layer from a digital file. The below provides summary information on how 3D printing works, the different types of technology involved, how it is used in various industries, and how you can use 3D printing for your own projects.

How 3D printing works

3D printing is an additive process rather than a subtractive process. For example, a sculptor carving a statue is a subtractive process—the sculptor starts with a big block of stone and chips away material until the stone resembles the desired figure. In an additive process such as 3D printing, a 3D printer starts with nothing and adds material layer by layer to build the object you wish to create. A layer is a thin cross-section of the object you are building, and each layer is attached seamlessly to the layer beneath it until the whole object is built. The 3D printer builds upon these layers based on instructions that it reads from a digital file.

The digital file for your 3D printer is exported from a computer-aided design (CAD) model, a digital blueprint of the desired object. You can either use special 3D modeling software or scan an existing object to create your 3D design. The design model is converted into a digital file comprising hundreds or thousands of horizontal slices. The 3D printer takes the specifications for each slice and transforms your desired material—plastic, gold, or even chocolate—into the layers of your object.

3D printing technologies

Three primary technologies used in 3D printing are stereolithography (SLA), fused deposition modeling (FDM), and selective laser sintering (SLS):

- SLA involves concentrating a beam of UV light on a vat of curable photopolymer resin. When touched by the light, the resin hardens and forms a single layer of the desired 3D object. The hardened layer is pulled out of the resin, creating space for more uncured resin, which is then exposed to light to create another layer. SLA is often used for creating medical models and prototyping product parts.

- FDM builds an object using material extrusion. Each layer of the object is created by selectively depositing melted material, usually thermoplastic polymers in filament form. FDM is a popular and well-known printing technology that is used in a variety of industries, including consumer goods and automotive manufacturing.

- SLS uses a high-powered laser to fuse, or "sinter," plastic, metal, ceramic, or glass powders to create layers for your 3D object. SLS is mainly used in the creation of prototypes and for low-volume production of component parts.

Industries that use 3D printing

As mentioned in the section on 3D printing technologies, many companies use 3D printing to build prototypes, models of their products that are still in development. For example, Nike uses 3D printing to print prototypes of new shoe designs using materials similar to those that will be used to produce the actual sneakers. The prototype allows them to rapidly test and make changes to the product before

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moving the shoe design to the production stage, potentially saving them thousands of dollars and many weeks of time. Automakers also use 3D printing to create prototypes—a recent innovation came from Local Motors, which created an entire car in just six days using a 3D printer.

While often it is used by companies for prototyping, 3D printing can also be used for the production of consumer goods. It is a popular production method for everyday goods such as jewelry, some kinds of clothing, and musical instruments including guitars and drum kits. Other uses for 3D printing include the production of replacement parts for automobiles and the creation of 3D models of priceless fossils for use in schools and museums.

In addition to consumer goods, 3D printing can prove useful within industries such as medicine or architecture. Within the construction industry, architects can use this technology to create 3D models of buildings and perhaps even the structure itself. For example, Winsun, a construction company based in China, has developed a continuous 3D printer for construction and printed its first batch of 10 houses in 2013. The medical industry has increased demand for 3D printing technology for creation of medical implants and artificial limbs. In the future, it may become possible to print human tissue and, someday, entire organs.

3D printing for personal use

In recent years, 3D printers have become more affordable, allowing artists, bakers, crafters, and other hobbyists immediate access to this versatile tool. RepRap was the first of the low-cost 3D printers and sells printer kits for around $1,000. However, if you don't wish to purchase your own 3D printer, you can still produce your own 3D-printed objects. There are several companies that let you upload your CAD models, and they will print your object and send it to you. Also, if you don't have access to commercial-grade 3D design software, Google has freeware for creating CAD models for personal projects. The technology's ability to produce three-dimensional objects quickly and cheaply has made 3D printing a popular do-it-yourself option for home projects.

3D Printing

A 3D printer is a device that enables you to create a three-dimensional object by creating it layer by layer. A layer is a thin cross-section of the object you are building. Each layer is attached seamlessly to the layer beneath it until the whole object is built. Even though there are different kinds of 3D printers, they all work the same way in that you need to begin with a computer-aided design (CAD) file, which provides the blueprint that the 3D printer uses to create the object. You can either create this CAD file using special 3D modeling software, or you can scan an actual object and create the CAD file that way. Then the software in the 3D printer "slices" the design into really thin horizontal layers, like a meat slicer does to a thick ham at the supermarket. There can be hundreds of these slices, or even thousands. Then the 3D printer "reads" the exact specifications of each slice and transforms the material into the exact specifications of the slice. The materials can be plastic, gold, other materials—even chocolate!

When a sculptor carves a statue, he is using what is called a subtractive process. This means he starts with a big block of stone or marble or wood and cuts away material until what is left looks like the thing he is carving, such as a bear. The industrial versions of this are cutting, milling, and machining. When a 3D printer is used, it is using an additive process. This means that the person starts with nothing, then adds layer after layer (using the instructions from the software) to build up the object you are creating.

If you don't own a 3D printer, you can still use one. There are several companies that let you upload your CAD file and they will "print" the object for you and send it back to you.

At Nike, they use 3D printing to "print" prototypes of new shoe designs using materials very much like the ones that will make it into the sneakers. If they don't like the way the prototype turns out, they can change the design in the software and print a new prototype. This way, they save thousands of dollars and many weeks of time. Other companies use 3D printing not to make prototypes but actual objects, such as jewelry and some kinds of clothing, that people buy. There is a company called RepRap which sells 3D printer kits that let you assemble your own 3D printer for around $1,000. (One day, you might be able to buy a kit, assemble the 3D printer, then have it create more 3D printers!) Google has some freeware for creating CAD files so you don't even have to buy expensive commercial-grade software. One technology used in 3D printing is called stereolithography (SLA). The way it works is that a beam of UV light is concentrated on a vat of photopolymer resin, making it soft. When a bead of that resin is drawn out of the vat it hardens in the light, creating the slice. Another technology used is called Fused Deposition Modeling (FDM). FDM uses extrusion, pushing a beam of melted thermoplastics into a layer, then adding another layer. FDM is used in such products as LEGOs that are made of ABS plastic or biodegradable polymer. Another method is called Selective Laser Sintering (SLS). SLS is like SLA but instead of using liquid in a vat it uses powders such as glass, nylon, ceramics, or metals. The powder is fused (the technical term is "sintered").

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3D printing is now used in medicine, where researchers are starting to create human tissue and, eventually, entire organs. Pharmaceutical companies are also using it, as are architects who are creating 3D models of buildings without having to make them out of cardboard or other materials. Artists were among the first people to see the potential for 3D printing, turning CAD files into unique, beautiful objects out of all kinds of plastics and metals. Archaeologists are creating 3D models of priceless fossils for use in schools and museums. There is even a 3D replica of King Tut's tomb!

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